

AI Career Assistance Portal

Product Submission Document

Version 1.0

Date: July 29, 2026

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Repository: <https://github.com/satyendramaurya123/Career-Assistance-Portal>

Status: Deployed — Live in <https://career-assistance-portal-wheat.vercel.app>

Submitted as fulfillment of the Product Requirements Document (PRD) v1.0

1. Executive Summary

AI Career Assistance Portal, originally scoped in the Product Requirements Document (PRD v1.0, July 6, 2026), has been fully designed, built, tested, and deployed to production. This document serves as the Product Submission Report: it confirms delivery against the original PRD, provides visual evidence (screenshots) of every major feature in its live, working state, records the actual technology stack used, and lists the resources and references relied upon during development.

The platform is a full-stack, AI-powered career preparation web application that allows job seekers to upload a resume and receive instant ATS scoring, job-match analysis, skill-gap detection, a personalized learning roadmap, AI-generated mock interviews, tailored cover letters, and LinkedIn profile content — all from a single authenticated dashboard.

1.1 Project Status

Development	Complete
Database & Auth (Supabase)	Complete
AI Integration (Gemini)	Complete
Deployment (Vercel)	Live
Google OAuth Sign-In	Complete

2. Objectives — Delivery Against PRD

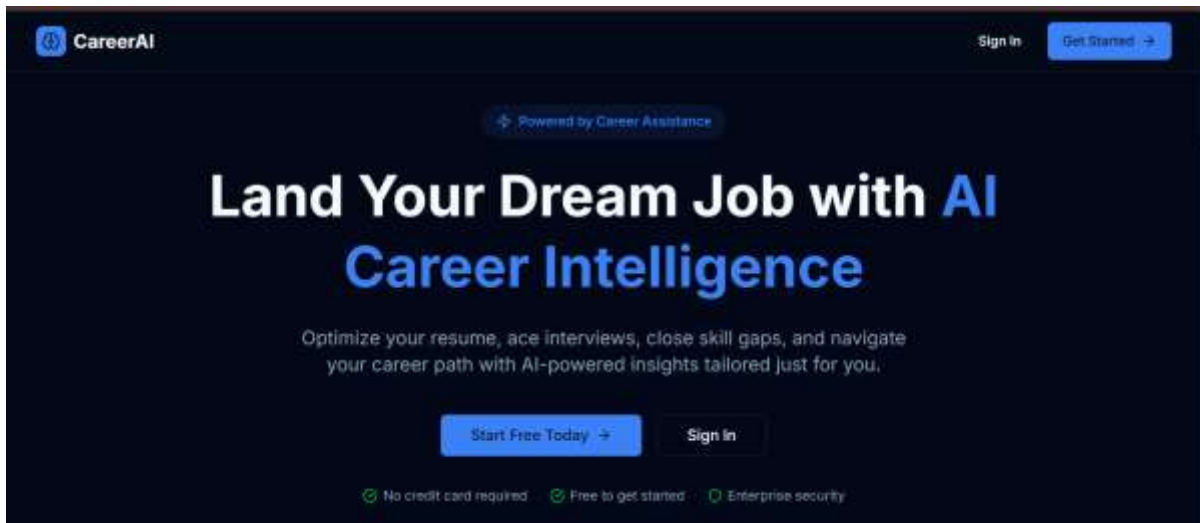
Every objective defined in the PRD has been met by the shipped product:

ID	Feature	Status
Automate Resume Evaluation	Instant, objective ATS compatibility score generated the moment a resume is uploaded, via Gemini-powered analysis.	Implemented
Close the Job-Fit Gap	Job Match module compares resume against a target role and computes a compatibility percentage with missing/recommended skills.	Implemented
Guide Skill Development	Skill Gap Analysis feeds directly into the AI Roadmap Generator, producing a personalized weekly/monthly learning plan.	Implemented
Build Interview Confidence	AI Mock Interview generates role-specific HR/Technical/Behavioral/System-Design questions and evaluates answers with scored feedback.	Implemented
Provide Secure, Role-Based Access	Supabase Auth (email/password + Google OAuth) with Row Level Security and an Admin role for platform oversight.	Implemented
Centralize the Journey	Unified Dashboard surfaces resume count, average ATS score, best job match, and score trends in one view.	Implemented

3. Features Delivered — Visual Evidence

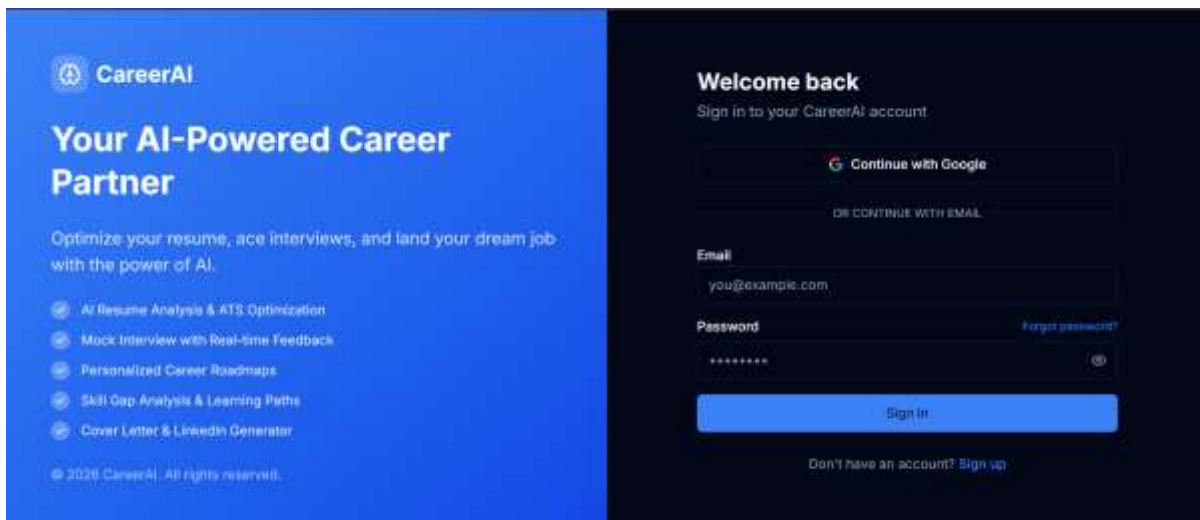
The following screenshots were captured from the live production deployment and demonstrate each PRD-defined feature (F1–F9) functioning end-to-end.

3.1 Landing Page & Onboarding



Public landing page — product pitch and call-to-action

3.2 F1 — Secure Authentication



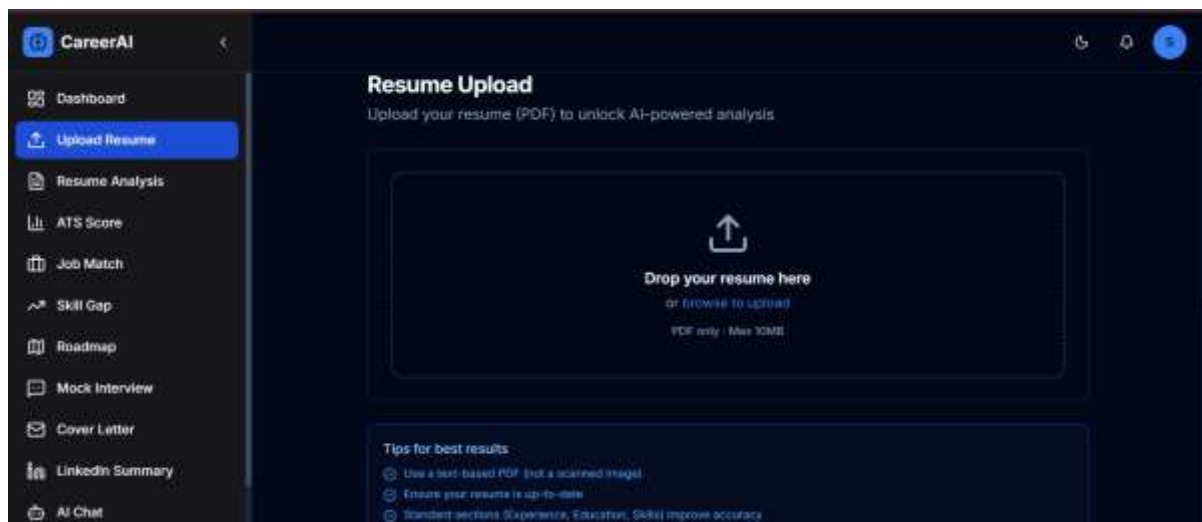
Login screen with Email/Password and “Continue with Google” (OAuth) options

3.3 F8 — Dashboard



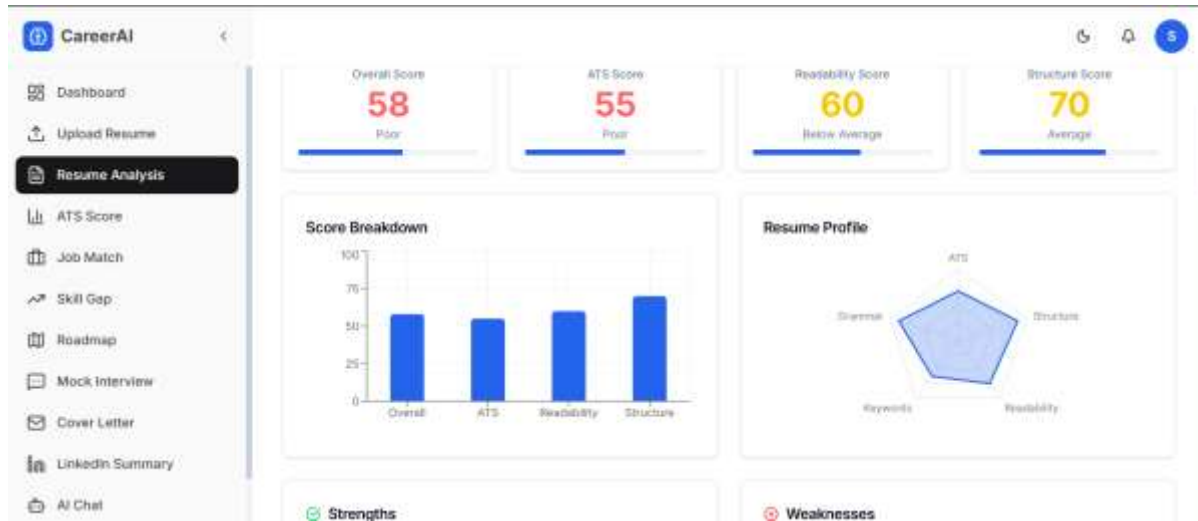
Unified dashboard — resumes uploaded, average ATS score, best job match, and score trend chart

3.4 F2 — Resume Upload & Parsing



Resume upload interface with drag-and-drop, PDF validation, and guidance tips

3.5 F3 — Resume Analyzer & ATS Score



AI-generated resume analysis — overall/ATS/readability/structure scores with breakdown chart and radar profile

3.6 F4 — Job Match

Job Match

See how well your resume matches your target role

Match Configuration

Enter your target role details for an AI-powered match analysis

Target Role *

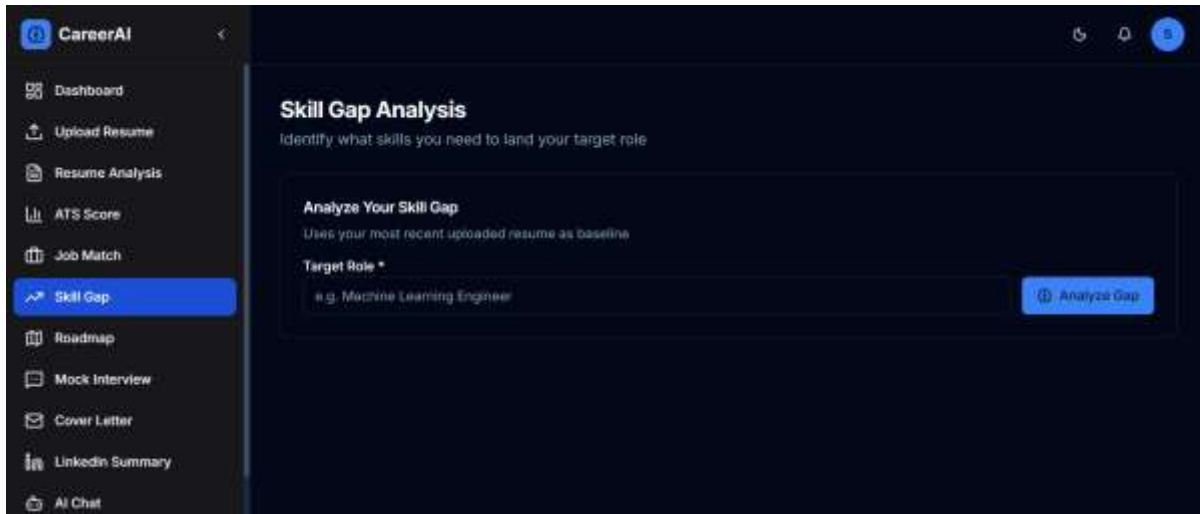
Experience Level *

Location (optional)

Job Description (optional)

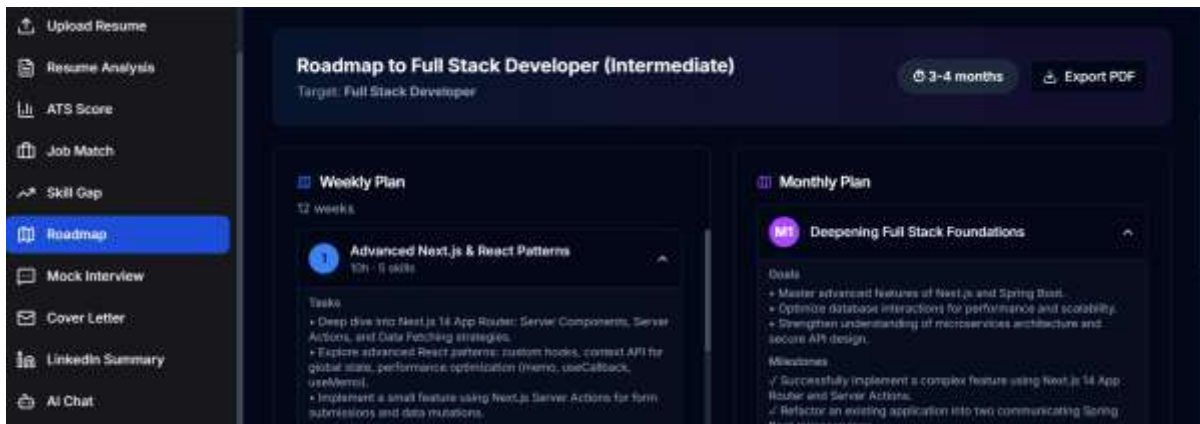
Job Match configuration — target role, experience level, location, and job description input

3.7 F4 — Skill Gap Analysis



Skill Gap Analysis — identifies missing/underrepresented skills against a target role

3.8 F5 — AI Roadmap Generator



Personalized learning roadmap with weekly task breakdown and monthly milestones, plus PDF export

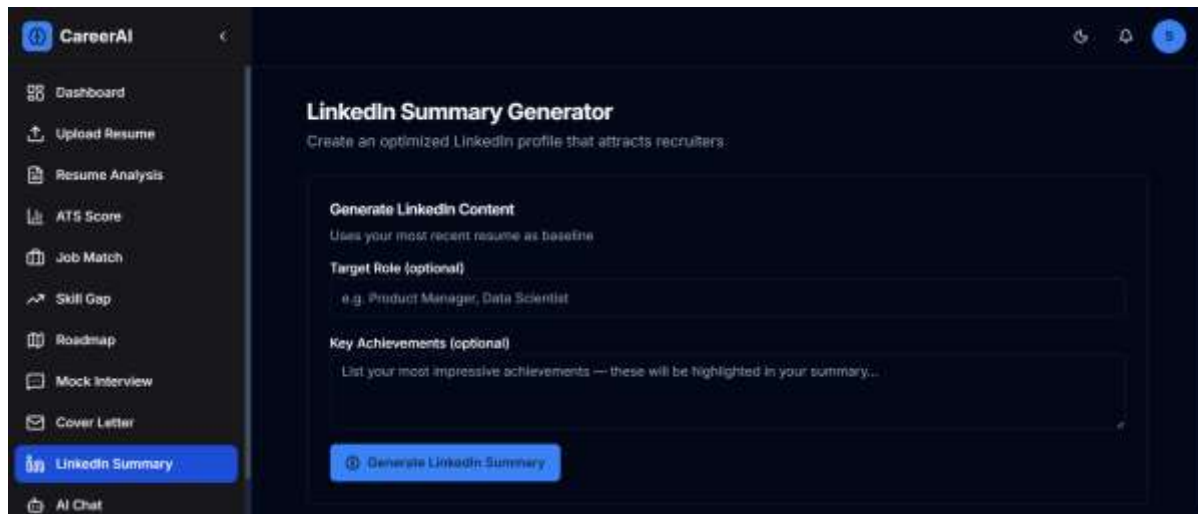
3.9 F6 — AI Mock Interview

Mock interview setup — target role, interview type (HR/Technical/Behavioral/System Design), and question count

3.10 F7 — AI Feature Suite: Cover Letter Generator

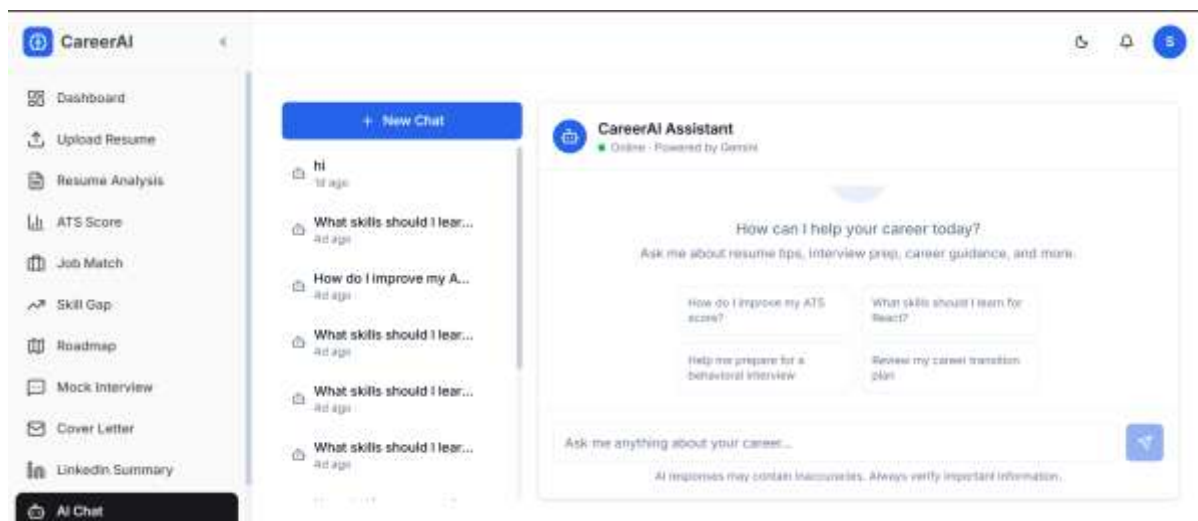
Tailored cover letter generation from job title, company, tone, and job description

3.11 F7 — AI Feature Suite: LinkedIn Summary Generator



AI-generated LinkedIn headline, about section, skills, and recruiter keywords

3.12 F7 — AI Feature Suite: AI Career Chatbot



Context-aware career chatbot with chat history, streaming responses, and quick-start prompts

3.13 F9 — Admin Panel

A role-gated Admin Panel is implemented (accessible only to users with the ADMIN role, enforced via middleware and Row Level Security). It surfaces total/active users, uploaded resumes, generated reports, AI request volume, and interview sessions, and allows admins to view, suspend, or delete user accounts, with all actions recorded to an audit log (admin_logs table).

4. Functional Requirements — Traceability

Mapping of each PRD functional requirement to its implementation status in the delivered product:

ID	Feature	Requirement	Status
FR-01	User Registration	Register with email and password	Implemented
FR-02	User Login	Validate credentials, issue session on success	Implemented
FR-03	Password Security	Passwords securely hashed before storage	Implemented
FR-04	Resume Upload	Accept PDF uploads, parse structured content	Implemented
FR-05	ATS Scoring	Generate ATS compatibility score per resume	Implemented
FR-06	Job Match Calculation	Compute match % between resume and target role	Implemented
FR-07	Skill Gap Analysis	Identify and rank missing/underrepresented skills	Implemented
FR-08	AI Roadmap Generation	Generate personalized learning roadmap	Implemented
FR-09	AI Mock Interview	Role-specific questions with response feedback	Implemented
FR-10	Cover Letter Generation	Tailored cover letter from resume + job description	Implemented
FR-11	Skill Recommendations	Suggest relevant skills based on role/trends	Implemented
FR-12	Dashboard View	ATS score, job match, skill gaps, roadmap progress in one view	Implemented
FR-13	Session Persistence	Session token persisted, re-authenticates on refresh	Implemented
FR-14	Admin User Management	View and manage registered user accounts	Implemented
FR-15	Admin Content Management	Edit AI prompt templates and question banks	Partial
FR-16	Logout	Invalidate local session on logout	Implemented

Note on FR-15: prompt templates are currently managed directly in the codebase (lib/gemini/prompts.ts) rather than through an in-app admin UI; this is flagged as a future enhancement in Section 7.

5. Technology Stack — As Delivered

The PRD's proposed stack (React/Vite, Express.js, JWT) evolved during development to a more integrated, production-ready set of managed services better suited to rapid, secure delivery:

Layer	Technology Used	Purpose
Framework	Next.js 15/16 (App Router) + TypeScript	Full-stack React framework — UI, API routes, and server actions in one codebase
Styling / UI	Tailwind CSS + ShadCN UI + Framer Motion	Utility-first responsive design system with accessible, reusable components
State Management	Zustand	Lightweight client-side state (resumes, chat, UI, auth)
Forms & Validation	React Hook Form + Zod	Type-safe form handling and schema validation
Database	Supabase (PostgreSQL)	Managed Postgres — 15 tables with foreign keys, indexes, soft deletes
Authentication	Supabase Auth (Email/Password + Google OAuth)	Secure session management, replacing custom JWT/bcrypt implementation
Authorization	Row Level Security (RLS) + role-based middleware	Per-row database access control, USER/ADMIN role separation
File Storage	Supabase Storage	Private “resumes” bucket (PDF, 10MB limit) and public “avatars” bucket
AI / LLM	Google Gemini API (@google/generative-ai)	Powers resume analysis, ATS scoring, roadmap, interviews, cover letters, chat
PDF Text Extraction	pdf-parse (pdfjs-dist)	Extracts resume text server-side for AI analysis
PDF Generation	pdf-lib	Generates styled, downloadable PDF reports (analysis, ATS, roadmap, etc.)
DOCX Generation	docx (npm)	Generates downloadable .docx cover letters
Charts	Recharts	Score trends, breakdowns, and radar profiles on the dashboard
Notifications	Sonner	Toast notifications across the app
Deployment / Hosting	Vercel	Serverless hosting with CI/CD from GitHub
Version Control	Git / GitHub	Source control and deployment trigger

5.1 Architecture Diagram (Logical, as Delivered)

Browser Client (Next.js UI) → HTTPS → Next.js API Routes / Server Actions → [Supabase Auth | Supabase Postgres + RLS | Supabase Storage | Google Gemini API] → response rendered back to the Dashboard.

6. Deployment & Access

Live Application: <https://career-assistance-portal-wheat.vercel.app>

Source Repository: <https://github.com/satyendramaurya123/Career-Assistance-Portal>

Hosting: Vercel (automatic deployment on every push to the main branch). Database, authentication, and file storage are provisioned via Supabase. Environment secrets (API keys, database credentials) are managed through Vercel's encrypted environment variables and are never committed to source control.

7. Known Limitations & Future Enhancements

- Gemini free-tier API quota is limited (requests/day); a paid billing tier would remove this constraint for higher-volume usage.
- Admin-side editing of AI prompt templates and interview question banks (FR-15) is currently code-based rather than exposed through the Admin Panel UI.
- SMS/Phone OTP authentication was scoped but not implemented (requires a paid third-party SMS provider); Email OTP was prioritized as the free, lower-friction option.
- Gemini model identifiers change frequently; the model name is externalized to an environment variable (GEMINI_MODEL) so it can be updated without a code deployment.

8. References

8.1 Project Links

Live Application: <https://career-assistance-portal-wheat.vercel.app>

Source Repository (GitHub): <https://github.com/satyendramaurya123/Career-Assistance-Portal>

8.2 Technology Documentation

Next.js Documentation: <https://nextjs.org/docs>

Supabase Documentation: <https://supabase.com/docs>

Supabase Auth (OAuth & Email OTP): <https://supabase.com/docs/guides/auth>

Google Gemini API Documentation: <https://ai.google.dev/gemini-api/docs>

Gemini Model Deprecations & Rate Limits: <https://ai.google.dev/gemini-api/docs/rate-limits>

Tailwind CSS Documentation: <https://tailwindcss.com/docs>

ShadCN UI: <https://ui.shadcn.com>

Vercel Deployment Documentation: <https://vercel.com/docs>

pdf-lib (PDF generation): <https://pdf-lib.js.org>

docx (DOCX generation, npm): <https://docx.js.org>

Zod (schema validation): <https://zod.dev>

React Hook Form: <https://react-hook-form.com>